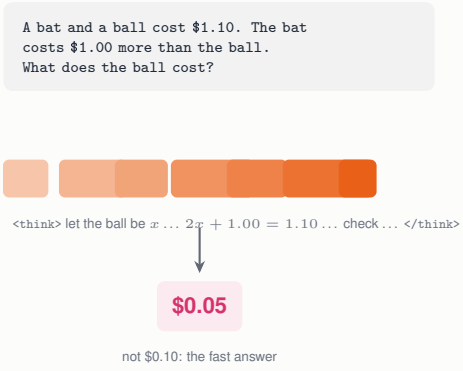
LECTURE NOTES · MODELS, 2026

Reasoning Models

Models that think before they answer: what the extra tokens buy, how chains are searched and verified, how reinforcement learning on checkable answers trains them (STaR to GRPO to DeepSeek-R1), distillation, test-time budgets, the 2026 landscape, and where the chains break.



A reasoning model spends tokens before it answers

Same transformer, new contract: a thinking span, trained for by reinforcement learning, under a budget.

a chat model answers

```
user: A bat and a ball cost $1.10. The bat costs
    $1.00 more than the ball. What does the ball cost?
assistant: The ball costs $0.10.
```

a reasoning model answers

```
assistant: <think>
Let the ball cost x. Then the bat costs x + 1.00,
so x + (x + 1.00) = 1.10, 2x = 0.10, x = 0.05.
Wait, check: 0.05 + 1.05 = 1.10. Yes. The common
mistake is to read "$1.00 more" as "the bat is $1.00".
</think>
The ball costs $0.05 (and the bat $1.05).
```

the span between the tags is generated like any other text, billed as output tokens, and in most products hidden or summarized; the answer after it is what the user sees

What is different

A thinking span. Hundreds to tens of thousands of tokens of intermediate work (decomposition, checking, backtracking) before the answer. Every token is one more forward pass: serial computation the single pass does not have (slide 6).

Trained for it. Not prompted: fine-tuned on long chains and then optimized by reinforcement learning against a checkable reward (slides 11–17), which makes the chains long, self-correcting and, on math and code, far more accurate.

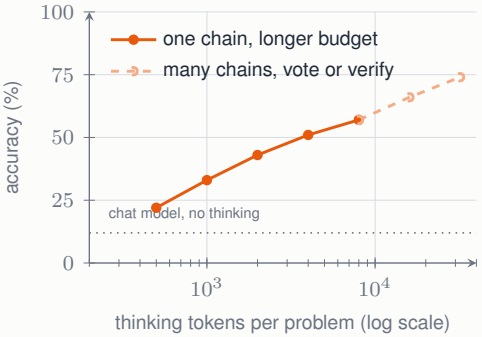
A budget. Accuracy rises with the tokens spent, so the caller chooses an effort level or a token budget (slide 19), and pays for it: the chain is output at output prices.

Three dials: training compute, test-time compute, and the data (verifiable problems). This lecture is about all three.

Why it works: accuracy scales with tokens spent at inference

A second scaling axis: for the same weights, thinking longer, or sampling more, buys accuracy on hard problems.

competition math (illustrative shape, after s1 and o1)



12 → 74%
AIME 2024, GPT-4o to o1†

83 / 93%
o1, 64 votes / 1,000 re-ranked†

Two axes of test-time compute

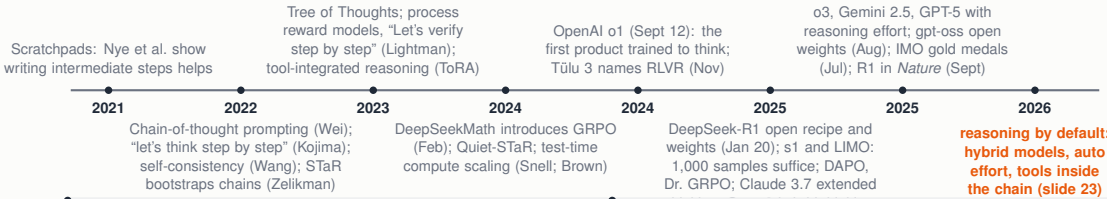
Serial: one chain, more tokens. The model revises, checks and backtracks inside it; the curve bends but keeps rising into the tens of thousands of tokens (s1, R1).

Parallel: many chains, then majority vote or a verifier. Coverage (pass@k) grows roughly log-linearly in k (Brown et al. 2024); picking the right one is the hard part (slide 8).

The price: the chain is billed as output: a 15k-token answer at $10 per million output tokens is $0.15 (derived), and an agent thinks at every step.

A short history: from scratchpads to reinforcement learning

Four years from “let’s think step by step” to models trained to think; the ideas arrived in a definite order.



What the order tells you

Prompting came first and showed the effect; verifiers and search showed that more compute at inference helps if you can pick the winner; RL on checkable answers removed the need to pick, by making one long chain good enough; distillation then made it cheap to copy. Each step is a slide in this deck.

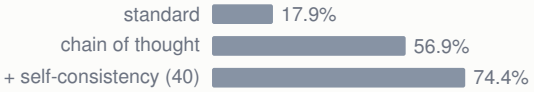
What stayed the same

The architecture. Every model on this timeline is a decoder-only transformer from the first deck, generating one token at a time. A reasoning model is a training recipe and an inference budget, not a new network, which is why open reproductions (slides 12, 18, 24) followed the first product within months.

The prompting era: ask for the steps and the answer improves

Few-shot exemplars with worked steps, a single magic sentence, and majority voting over sampled chains.

GSM8K accuracy, PaLM 540B (Wei et al. 2022; Wang et al. 2022, whose own CoT baseline is 56.5)



Three results that started it

Chain-of-thought prompting (Wei et al. 2022): put worked solutions in the few-shot exemplars and a large enough model writes its own; the gain appears above about 100B parameters.

Zero-shot CoT (Kojima et al. 2022): the sentence "Let's think step by step" alone takes InstructGPT from 17.7% to 78.7% on MultiArith and 10.4% to 40.7% on GSM8K.

Self-consistency (Wang et al. 2022): sample many chains, take the majority answer; the first test-time scaling result.

What the era established, and what it could not do

Established: the steps are the mechanism, not decoration (remove them and the accuracy goes with them); longer, more structured chains help on harder problems; sampling and voting buy accuracy with compute.

Could not do: make the model *check* itself. Huang et al. (2023): asked to review its own answer without external feedback, a model's accuracy on reasoning benchmarks fell rather than rose. The chain was the user's trick, not the model's habit.

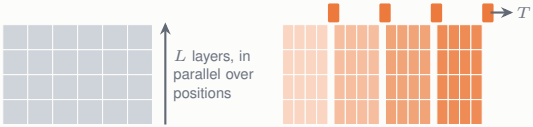
Bootstrapping: STaR (Zelikman et al. 2022) kept the chains that led to correct answers and fine-tuned on them: the seed of every later recipe.

Scratchpads (Nye et al. 2021) and **least-to-most** decomposition (Zhou et al. 2022): the same idea for code and for splitting problems.

What the extra tokens buy: serial computation

One forward pass is a fixed-depth circuit; each generated token is another pass, so a chain of T tokens has depth $L\,T$.

one forward pass over n tokens T passes, each appending one token



depth L whatever the problem: a log-precision transformer sits in the circuit class TC^0 (Merrill & Sabharwal 2023)

each new token is written to the context and read by the next pass: a scratch cell; depth $L \times T$, with T chosen at inference

What the theory says

Feng et al. (2023): bounded-depth transformers cannot evaluate growing arithmetic expressions or linear systems directly; a constant-size one can with a chain of thought. **Merrill & Sabharwal (2024):** linearly many steps reach regular languages the single pass cannot; polynomially many reach all of P. **Li et al. (2024):** T chain tokens let a constant-depth transformer simulate a circuit of size T.

What it does and does not explain

Explains: why steps help on multi-step problems (long arithmetic, compositions, search), why longer chains help harder problems, and why a bigger model without a chain is not a substitute: width buys parallel work, tokens buy serial work.

Does not explain: which chain the model writes, why sampling finds good ones, or why RL makes them reliable. The results are about existence; the mechanism in trained models is in-context search: propose, check, backtrack (slides 12, 15).

Alternatives: the scratch cell need not be a word. Coconut (Hao et al. 2024) feeds the last hidden state back as a continuous “thought”; recurrent-depth models (Geiping et al. 2025) loop layers instead of emitting tokens: the same serial compute, no text to read or monitor.

Is the chain the reason? Faithfulness and monitoring

The text often omits what moved the answer; it is still our best window, and optimizing it closes the window.

Three measurements

Turpin et al. (2023): reorder few-shot exemplars so the answer is always (A), and accuracy drops by up to 36 points while the chains argue for (A) without ever mentioning the pattern.

Lanham et al. (2023): truncate or corrupt the chain and see whether the answer changes; often it does not.

Chen et al. (2025): give a reasoning model a hint that changes its answer; Claude 3.7 Sonnet mentions the hint in 25% of those chains, DeepSeek-R1 in 39%; an RL-trained Claude 3.7 snapshot exploits reward hacks in over 99% of episodes and says so in under 2% (five of six environments).

Monitoring, and the trade-off

Baker et al. (2025): a weaker model reading the chain catches a frontier coding agent's reward hacks ("let's hack") better than reading its actions; penalizing bad thoughts during training taught the agent to hack while hiding the intent.

Korbak et al. (2025), a cross-lab position paper: chain-of-thought monitorability is "a new and fragile opportunity"; track it, and weigh it before training choices that could erode it.

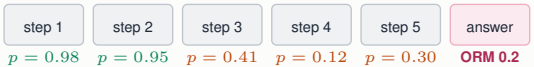
In products the raw chain is usually hidden or summarized (OpenAI, Anthropic), sometimes shown (DeepSeek).

How to hold it: the chain is evidence, not proof. Slide 6 says it is where the serial computation happens, so it is *necessary* on hard problems; this slide says it is not a complete account of *why*. Monitor it, and never optimize it for appearance.

Process or outcome: what should a verifier grade?

A reward model can score the final answer or every step; step-level supervision won in 2023 and lost to scale in 2025.

one sampled solution to a MATH problem



a process reward model (PRM) scores every step, an outcome reward model (ORM) the whole; step 3 is where this solution went wrong, and only the PRM says so

The 2023 result

Lightman et al. (2023), *Let's verify step by step*: a PRM trained on 800k human step labels (PRM800K) picks the right solution out of 1,860 samples 78.2% of the time on a MATH subset, against 72.4% for an ORM and 69.6% for majority vote. Uesato et al. (2022) had found the two close on GSM8K; Math-Shepherd (Wang et al. 2023) replaced human labels with Monte Carlo rollouts.

Why the trained models went the other way

DeepSeek-R1's report lists PRMs among its *unsuccessful attempts*: defining a step is hard in general reasoning, judging a step is hard even for humans, and a learned PRM invites reward hacking and must be retrained as the policy moves. Monte Carlo tree search over steps failed too: the token space is too large and the value model too hard to train.

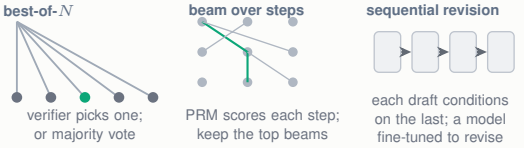
What survived: outcome rewards that a program can check (an answer string, a unit test) and the discipline of Gao et al. (2022): any learned reward is over-optimized if you push on it long enough. Generative verifiers (GenRM 2024; DeepSeek-GRM 2025) and rubric rewards are the 2025 attempt to grade non-verifiable work without a brittle scalar.

PRMs live on as selectors at inference (slide 9).

Search at inference: sample, select, revise

Before the chain was trained, compute was spent around a fixed model: samples, a verifier to pick, a tree to explore.

three ways to spend N samples



Compute-optimal scaling (Snell et al. 2024)

On MATH with PaLM 2-S*, the best strategy depends on difficulty: revisions win on easy problems, search with a PRM on hard ones; allocating adaptively is about 4× more efficient than best-of-N, and in a FLOPs-matched comparison test-time compute beats a 14× larger model on easy and medium problems, not on the hardest.

Coverage: the monkeys result

Brown et al. (2024): the fraction of problems solved by at least one of N samples (coverage) grows roughly log-linearly in N for four orders of magnitude; on SWE-bench Lite, DeepSeek-Coder-V2 goes from 15.9% with one sample to 56% with 250, past the single-sample state of the art. The catch is identification: with no verifier, majority voting and reward models plateau beyond a few hundred samples while coverage keeps rising.

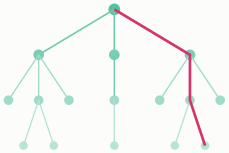
Tree of Thoughts (Yao et al. 2023) made the search explicit, with the model proposing and evaluating partial thoughts.

What this bought: a way to turn compute into accuracy with a frozen model, and the question o1 answered: can the model learn to do the search *inside one chain*?

From search around the model to search inside the chain

o1's bet: train the model so that one long chain does the proposing, checking and backtracking that trees did outside.

search outside: a frozen model, a tree, a verifier



a PRM or vote picks the path; the model never sees its own dead ends

search inside: one chain that proposes, checks and backtracks



the dead end stays in the context and conditions what comes next; the verifier is the model's own check; the budget is the search depth

What the trained chain does better

No selector needed: the chain reaches one answer; R1-Zero 71.0% pass@1 on AIME 2024, and voting over 64 chains still adds (86.7%), so the two axes compose.

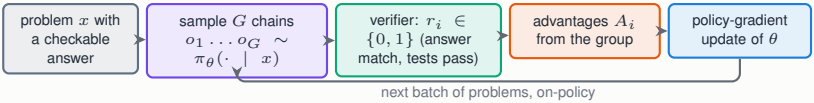
Dead ends are information: a tree discards a failed branch; the chain keeps it in context and avoids it. o1's report[†]: the model "learns to recognize and correct its mistakes."

Why the trees were abandoned: the R1 report lists MCTS over tokens and process reward models among its unsuccessful attempts: a search space too large to tile with a value model, and a learned step grader that the policy games (slide 8).

What it costs: the search is now serial: latency and tokens grow with difficulty, capped only by the budget (slide 19).

RL with verifiable rewards: the chain becomes the policy

Sample whole chains, check the final answer with a program, push probability toward the chains that were right.



$$J(\theta) = \mathbb{E}_{x\sim D,\; y\sim\pi_\theta(\cdot|x)}\big[\, r(x,y) \,\big] - \beta\, \mathrm{KL}\big(\pi_\theta \,\|\, \pi_{\mathrm{ref}}\big)$$

$$\nabla_\theta J = \mathbb{E}\Big[\sum_t \nabla_\theta \log \pi_\theta(y_t \mid x, y_{<t})\; A_t\Big]$$

state: prompt plus tokens so far; action: the next token; episode: one whole chain and answer; reward: once, at the end. A_t is the advantage, the reward minus a baseline; the baseline is where the algorithms differ (slide 13).

Why this works where RLHF stumbles

The reward is a program, not a learned model: a boxed answer, a unit-test suite, a compiler. It cannot be flattered, does not drift, and is free; Tülu 3 (Lambert et al. 2024) named the setting RLVR.

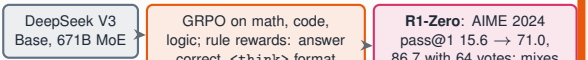
The signal is sparse and honest: one bit per chain, so the policy may find *any* chain that works; length, verification and backtracking are what survived, not what was prescribed.

Rollouts are the cost: thousands of problems, 8–64 chains each, thousands of tokens each.

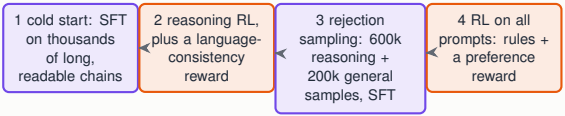
DeepSeek-R1: the open recipe, in two tracks

Pure RL from a base model shows the behavior; a four-stage pipeline with a cold start makes it a usable model.

R1-Zero: no supervised data at all (DeepSeek-AI, January 2025)



R1: four stages



stage 2 produces the chains stage 3 harvests; stage 4 restores the helpfulness and harmlessness that reasoning RL ignored. The 2025 *Nature* version puts the RL training (with its SFT data) at about $294K of H800 time at $2 per GPU-hour, on top of the base model[†].

Why the cold start matters: a few thousand curated chains fix format and language before RL amplifies them; R1-Zero shows what RL finds on its own, R1 what you ship.

What came out

R1: AIME 2024 79.8%, MATH-500 97.3%, GPQA Diamond 71.5%, Codeforces rating 2,029 (above 96% of humans), SWE-bench Verified 49.2%; weights under MIT license, chains shown to users.

Distillation beats RL for small models: 800k R1 samples fine-tune Qwen and Llama from 1.5B to 70B; the 32B student scores 72.6% on AIME against 47.0% for RL run directly on the same base (slide 17).

The lesson that generalized: a strong base, verifiable problems, a rule reward, a group baseline, and enough steps for the chains to grow (slide 15). Within weeks every open lab had a reproduction (slide 24).

GRPO and its descendants: a baseline from the group

Drop the value model, normalize each chain's reward against its siblings; then fix the biases that this introduces.

$$A_i = \frac{r_i - \mathrm{mean}(r_1, \ldots, r_G)}{\mathrm{std}(r_1, \ldots, r_G)} \qquad \text{(Shao et al. 2024)}$$

$$J = \mathbb{E}\Big[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_{t=1}^{|o_i|}\ell_{i,t}\Big] - \beta\,\mathrm{KL}(\pi_\theta \| \pi_{\mathrm{ref}})$$

$$\ell_{i,t} = \min\big(\rho_{i,t}A_i,\ \mathrm{clip}(\rho_{i,t}, 1-\varepsilon, 1+\varepsilon)A_i\big)$$

$\rho_{i,t} = \pi_\theta(o_{i,t} \mid \cdot)/\pi_{\theta_{\mathrm{old}}}(o_{i,t} \mid \cdot)$: importance ratio; G: group size, 8–64; ε: PPO clip; Shao et al. estimate the KL per token inside the sum

What the group buys

PPO needs a value network as large as the policy for its baseline; GRPO uses the mean reward of G siblings sampled for the same problem: half the memory, and no critic to train on sparse rewards. RLOO (Ahmadian et al. 2024): a leave-one-out mean, no clipping; REINFORCE++: PPO's clip, a global baseline. All REINFORCE variants.

The 2025 corrections

Length bias: the $1/|o_i|$ term and the std normalization make long wrong chains cheap and short right chains dear; Dr. GRPO (Liu et al. 2025) removes both, and the ever-growing length of R1-Zero-style runs with them.

DAPO (Yu et al. 2025): clip-higher ($\varepsilon_{\mathrm{high}} = 0.28$) so rare tokens can grow; dynamic sampling that drops all-equal groups (no gradient); a token-level loss; soft penalties for overlong chains. Qwen2.5-32B base reaches AIME 2024 50 against 47 for the R1-style baseline, in half the steps.

Sequence-level ratios: GSPO (Qwen 2025) stabilizes RL on mixture-of-experts models; CISPO (MiniMax-M1) clips the importance weights, not the token updates.

Entropy collapses fast; the high-entropy "fork" tokens (the top 20% by entropy) carry the useful gradient (Wang et al. 2025).

Rewards and verifiers: what is checked, what is gamed

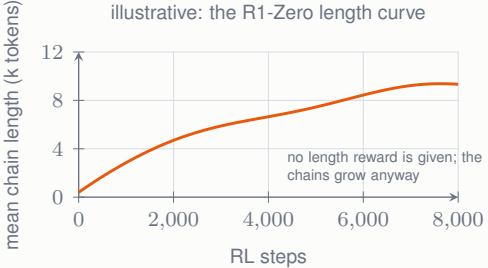
The recipe is only as good as the checker; every learned or soft reward is a target the policy will find a way around.

Reward	How it is computed	How it gets gamed, and the fix
Answer match	normalize the boxed answer (SymPy, string rules) and compare	loose normalization rewards wrong forms; multiple choice lets guessing score; use free-form answers and strict parsers
Unit tests	run the code in a sandbox against hidden tests	special-casing visible tests, editing the test file (Baker et al. 2025 caught "let's hack" in the chain); hidden tests, sandboxes, chain monitors
Format, length	tags present, one answer, target language; soft caps on chain length	language mixing and tag spam (R1 adds a language-consistency reward); padding or truncation (DAPO's soft overlong penalty)
Learned judge	an LLM or reward model scores quality	over-optimization (Gao et al. 2022); flattering the judge; generative verifiers, rotated judges
Rubric	a per-task checklist scored by a model	brittle wording; the best current option for non-verifiable domains

The boundary of the method. RLVR needs a checker, so it reached mathematics, code and formal tasks first, and that is where the 2025 gains are. Writing, research, advice and open-ended agent work get learned judges, rubrics and preference data, which bring back every RLHF failure; slide 21 shows how agentic recipes turn tool results into checkable outcomes.

What RL changes, and what was already there

Chains grow and start to check themselves; two results say much of that behavior was in the base model already.



Dynamics: accuracy and length rise together; entropy falls; the distribution narrows onto chains that work (slide 16). **And:** report a second model family and a random-reward control, or a small-scale RL claim means little.

What appears during training

R1-Zero's chains grow from hundreds of tokens to about ten thousand with no length reward, and acquire re-checking, backtracking and the "aha moment" the report quotes. Liu et al. (2025, Dr. GRPO): the base models (Qwen2.5, DeepSeek-V3-Base) already write such phrases before any RL, and part of the length growth is an artifact of GRPO's normalization (slide 13).

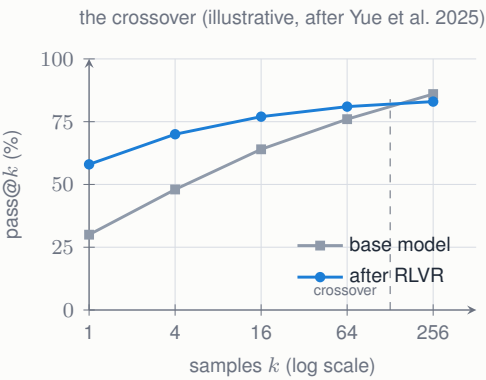
Two cautionary results

One example is enough (Wang et al. 2025): RLVR on a single problem takes Qwen2.5-Math-1.5B from 36.0% to 73.6% on MATH500.

Spurious rewards (Shao et al. 2025): on Qwen2.5-Math-7B, *random* rewards add 21.4 points on MATH500, wrong labels 24.1, the true reward 29.1; Llama and OLMo gain nothing.

Does RL create or select? Sharpening, coverage, entropy

RL makes the model find in one try what the base found in many; whether it finds anything new is the open question.



Three readings of the same curve

Sharpening (Yue et al. 2025): the RL model wins at small k and the base matches or beats it at large k (math, code, visual): RL raises the probability of chains the base could already produce and narrows coverage; distillation adds chains.

Discovery with more steps (ProRL, Liu et al. 2025): with thousands of steps, KL control and reference resets, a 1.5B model's pass@k passes its base where the base had none. Khatri et al. (2025), 400k GPU-hours: RL curves are sigmoids in compute; recipes differ in ceiling, most details only in efficiency; early slope does not predict the ceiling.

Entropy (Cui et al. 2025): policy entropy collapses early and performance follows a fixed curve of it; keep it up and the ceiling rises. Wang et al. (2025): the top 20% of tokens by entropy (the "forks") carry the useful gradient.

Distillation: a thousand good chains, and a word

Fine-tuning on a few hundred long chains plus a budget control recovers most of the behavior; the base model had it.

1,000
examples fine-tune s1-32B

817
examples fine-tune LIMO

“Wait”
appended to force more thinking

26 min
on 16 H100s for s1-32B

Budget forcing (s1, Muennighoff et al. 2025)

Fine-tune Qwen2.5-32B-Instruct on 1,000 questions with traces distilled from Gemini, chosen for difficulty, diversity and quality. At inference, cap the think span by inserting the end-of-thinking token, or extend it by suppressing that token and appending “Wait”: the model re-checks and often fixes its answer. s1-32B beats o1-preview on competition math by up to 27%.

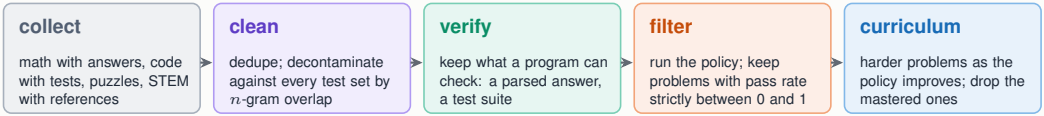
Less is more (LIMO, Ye et al. 2025)

817 curated problems with high-quality chains take Qwen2.5-32B-Instruct to 57.1% on AIME 2024 and 94.8% on MATH (v1), against 6.5% and 59.2% for earlier SFT recipes with 100k examples. The authors’ reading: the pretrained model holds the knowledge; a few exemplary chains teach the *format*. The caveat: the chains came from RL-trained models (slide 12).

Why it matters: if a thousand examples unlock long, self-checking chains in a 32B base model, the behavior is latent in pretraining; RL’s job is to find it and make it reliable where no exemplar exists (slide 16).

Where the problems and rewards come from

Verifiable problems are the scarce input; the pipeline is collection, cleaning, verification and difficulty filtering.

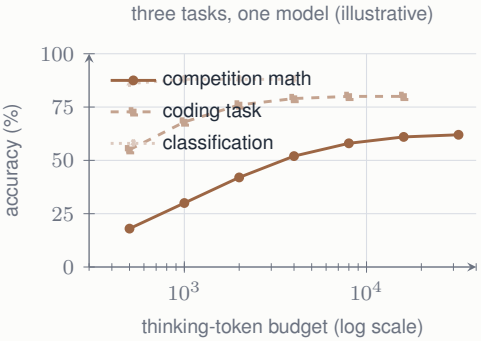


Source (open)	Size[†]	What it is
NuminaMath (2024)	860k	competition and olympiad problems with chain-of-thought solutions; the AIMO prize data
OpenR1-Math-220k (2025)	220k	R1-generated solutions to NuminaMath problems, verified; Hugging Face's R1 reproduction
OpenMathReasoning (2025)	540k problems, 3.2M solutions	NVIDIA's AIMO-2 winning data, long chains plus tool-use traces
DAPO-Math-17k (2025)	17k	integer-answer problems used to train DAPO; the small, clean RL set
OpenThoughts3 (2025)	1.2M	math, code and science traces for distillation; OpenThinker3 students
SWE-Gym (2024), SWE-smith (2025)	2.4k, 50k	executable repository tasks with tests for agentic coding RL

The limiting resource is the checker, not the text: math has a parser (Math-Verify), code has tests, everything else needs a rubric or a judge (slide 14). At the extreme, one problem (Wang et al. 2025) or self-proposed problems checked by a code executor (Absolute Zero) still train.

Budgets and effort: deciding how much to think

The caller sets the budget, the server closes the thinking span at the limit, accuracy follows and then saturates.



The knobs, as shipped[†]

Effort levels: OpenAI's `reasoning_effort` (none to max, model-dependent; low/medium/high on gpt-oss); Anthropic's adaptive thinking with `effort` (`budget_tokens` up to Claude 4.5); Google's `thinking_level` (`thinkingBudget` on Gemini 2.5); DeepSeek's thinking switch plus `reasoning_effort`; Qwen3's `/think` with a cap.

How a cap is enforced: the server counts thinking tokens and at the limit inserts the end-of-thinking token (s1's budget forcing); to think *more*, s1 suppressed it and appended "Wait".

Hybrid models (Qwen3, Claude, DeepSeek V3.1, GPT-5) answer with or without a chain from one set of weights; by 2026 the model often picks its own effort (slide 23).

Rule of thumb: start low; raise effort only where it pays.

Overthinking, underthinking, and learning the right length

Trained chains waste tokens on easy problems and give up too early on hard ones; length is now a trained quantity.

Two failure modes

Overthinking (Chen et al. 2024, “Do not think that much for 2+3”): o1-style models pile redundant solutions onto trivial problems; a trained preference for efficient chains halves MATH500 tokens at equal accuracy.

Underthinking (Wang et al. 2025, “Thoughts are all over the place”): on hard problems the models abandon promising lines early and switch; wrong answers switch more; a decoding penalty on switching raises accuracy without retraining.

Agents overthink too (Cuadron et al. 2025): deliberating instead of acting lowered SWE-bench success.

Making length a controlled variable

L1 (Aggarwal and Welleck 2025): RL with a target length in the prompt; the model obeys the target and, at equal length, beats budget forcing; long-chain models are strong *short* reasoners once trained to be.

NoThinking (Ma et al. 2025): prefill a dummy thinking block; at small budgets it beats thinking, and with parallel samples it matches thinking at a fraction of the latency.

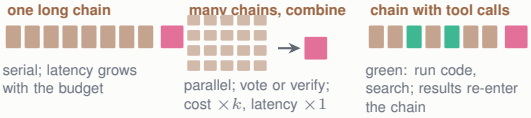
Adaptive thinking (AdaptThink, Thinkless 2025): RL that chooses whether to think at all, halving chain length at equal or better accuracy on easy sets. DAPO’s overlong shaping and Kimi k1.5’s length penalty do the same during training.

The picture to keep: an accuracy-against-tokens frontier per task. Budget forcing walks along one model’s curve; length-trained models move the curve left; adaptive models pick the point per question.

Parallel compute, and tools inside the chain

Beyond one longer chain: many chains combined, and chains that call a program or a search engine mid-thought.

three shapes of inference compute



Parallel, in products[†]

o1 reached 93% on AIME with 1,000 samples re-ranked; Gemini's Deep Think and Grok's Heavy mode sell parallel hypotheses with a synthesis step; OpenAI's "pro" tiers sell more compute per answer without stating the mechanism. Snell et al.: the right mix depends on difficulty; the catch is still picking the winner (slide 9).

Tools inside the chain

Tool-integrated reasoning (ToRA, Gou et al. 2023): interleave natural-language steps with program calls; arithmetic and algebra go to the interpreter.

Trained with RL: ReTool (Feng et al. 2025) puts a code interpreter in the RL loop; a 32B model reaches 67% on AIME 2024 in 400 steps where text-only RL took 1,080 steps for 40%. o3, GPT-5 and Claude 4 think between tool calls; Kimi K2 Thinking runs hundreds of sequential calls in one task[†].

Agentic RL: the verifier is the environment. SWE-RL (Wei et al. 2025) rewards patch similarity on real repositories; Qwen3 and GLM report RL on multi-turn tool use; chain, actions and observations form one trajectory, and every tool result re-prefills it (slide 22).

Serving a reasoning model: the chain is output

Thinking tokens are decoded one at a time, sit in the KV cache, and are billed; the runtime splits and caps them.

one answer (derived)	2k tokens	10k tokens	30k tokens
decode time at 60 tok/s	33 s	2.8 min	8.3 min
KV, 70B dense, 320 KB/token	0.6 GB	3.2 GB	9.6 GB
KV, V3-class MLA, 69 KB/token	0.14 GB	0.7 GB	2.1 GB
cost at $10 per M output tokens	$0.02	$0.10	$0.30
cost at $60 per M (o1, 2024)	$0.12	$0.60	$1.80

per-token figures from the runtime deck; list prices at the time[†]; reasoning tokens are billed as output everywhere

What the runtime does

Parse: the reasoning parser routes the think span to a separate field (`reasoning_content`) and the rest to the answer; clients can hide one and must keep the other.

Cap: the server closes the span at the budget. **Stream:** the chain or its summary goes out as its own delta.

What it changes for the operator

Decode-bound: a long chain is thousands of memory-bound steps; throughput comes from batching many thinking requests, and a single hard problem has a latency floor no hardware shortcut removes.

Memory: chains hold the cache for minutes; capacity planning is in token-minutes, not requests.

Across turns and tool calls: DeepSeek ignores a previous chain in plain chat but requires it back across tool calls; Anthropic keeps signed thinking blocks so the plan survives a tool call; either way the transcript grows and the prefix cache must span it.

Hidden chains still cost: a summarized chain is billed in full; a budget is the only lever on the bill.

October 2026: effort dials and always-on thinking

Two years after o1 every lab ships one; the knobs converged on effort levels, and the chain often cannot be turned off.

Model (released)	Org	Weights	Reasoning controls	$/M in / out	GPQA-D	HLE
GPT-6 Astra (Sep 3, 2026)	OpenAI	closed	effort low / medium / high / xhigh / max; hidden chain; 1M context	10 / 50	96.0	57.2 (tools)
GPT-5.6 Sol, Terra, Luna (Jul 9, 2026)	OpenAI	closed	API effort none to max; *ultra* (ChatGPT, Codex) runs four agents in parallel	4/20, 2/12, 0.2/1.2	94.6 (Sol)	—
Claude Opus 5.5 (Sep 22, 2026)	Anthropic	closed	adaptive thinking, effort low to max; cannot be switched off; signed thinking kept across tool calls	4 / 20	—	67.7 (tools)
Claude Fable 5.1 (Sep 1, 2026)	Anthropic	closed	adaptive, always on; summarized or omitted display, billed in full	10 / 50	—	65.0 (tools)
Gemini 4 Argon (Sep 30, 2026)	Google	closed	`thinking_level` low / medium / high; thought signatures; Deep Think retired into Argon	2/10 (4/20 from Dec)	—	—
DeepSeek V4-Pro (Apr 24, 2026)	DeepSeek	MIT	non-think / think high / think max; 1.6T MoE, 49B active, 1M context	1.32 / 3.96 peak	90.1	37.7 (48.2 tools)
Qwen3.8-2.4T-A95B (Aug 12, 2026)	Alibaba	custom	effort xhigh / medium / low; “thinking cannot be disabled”	self-hosted	92.6	43.6 (56.2 tools)
Kimi K3 (Jul 2026)	Moonshot	custom	effort low / high / max, always on; 2.8T MoE, 104B active	self-hosted	93.5	43.5 (56.0 tools)

vendor pages and model cards read Oct 21–22, 2026†; GPQA Diamond and Humanity’s Last Exam as each vendor reports them (Fable 5.1’s GPQA, 93.7, is third-party only), not comparable across rows; API list prices (Sol’s is promotional). Thinking tokens are billed as output everywhere, shown or not. Token budgets gave way to effort levels: `budget_tokens` is deprecated on current Claude models, `thinking_level` replaced `thinkingBudget`; gpt-oss (slide 24) exposes three levels in the open.

Open weights and open recipes

The recipe is public, the data is public, and the stacks are public; what is scarce is the base model and the GPU-hours.

Open reasoning models people actually pull†

Hugging Face downloads, 30 days to Sep 18, 2026 (one snapshot): Qwen3-0.6B 22.5M, Qwen3-8B 13.0M, Qwen3.6-35B-A3B-FP8 10.2M, Gemma 4 26B-A4B 9.8M, gpt-oss-20b 6.7M, DeepSeek V4-Flash 4.4M, Kimi K3 (a repack) 3.3M: small hybrids by count, trillion-parameter models at the ceiling. Licenses: MIT (DeepSeek), Apache-2.0 (gpt-oss, Gemma 4, Qwen3 to 27B), revenue-gated terms for Qwen3.8-2.4T and Kimi K3.

What a reproduction costs†

TinyZero: R1-Zero's countdown behavior on a 3B model for about $30; s1: 1,000 examples, 16 H100s, 26 minutes; DeepSeek's RL phase on top of V3: about $294K of H800 time (*Nature*, 2025); OpenThinker3-7B: supervised fine-tuning alone on 1.2M traces reaches 53% on AIME 2025.

Stacks, data, and what the labs disclose

Training stacks: verl, OpenRLHF, TRL's GRPO trainer, SkyRL, ROLL, slime, AReaL (asynchronous rollouts), NVIDIA NeMo RL, prime-rl: all implement GRPO-family algorithms with vLLM or SGLang as the rollout engine.

Data: OpenThoughts3 (1.2M traces; the authors found QwQ-32B a better teacher than R1), NVIDIA's 3T tokens of released Nemotron 3 data including RL sets, NuminaMath, OpenMathReasoning, DAPO's 17k (slide 18).

Recipes in cards: Kimi K3 trained nine specialists (three domains by three effort levels) and merged them by on-policy distillation, with a -1 reward for overshooting the budget; Qwen3's four-stage post-training. The frontier labs disclose effort levels and prices, not recipes; a report that GPT-6 Astra loops layers (recurrent depth, slide 6) is unconfirmed.

Benchmarks and competitions: saturation, harness sensitivity

The 2024 tests are solved; the field moved to agentic tasks, cost per task, and competitions graded by humans.

Competitions, with dates[†]

IMO 2025 (July): Gemini Deep Think 35 of 42, graded by IMO coordinators; an OpenAI experimental model 35 of 42, graded by former medallists. **IMO 2026 (July):** Xiaohongshu reported 42 of 42 graded by IMO organizers and Huawei claimed the same (company claims via AFP; no official AI results); in one community harness GPT-5.6 Sol scored 28 at default effort, 39 at xhigh.

ICPC 2025: OpenAI 12 of 12, Gemini 2.5 Deep Think 10 of 12. **ARC Prize 2025:** best open entry 24% on ARC-AGI-2 at $0.20 per task, Opus 4.5 37.6% at $2.20: cost is now an axis.

The honest reading: on closed-form math and code the models now exceed almost all humans; the open questions are cost, reliability, and problems without a checker.

What the numbers mean now

Saturation: MATH-500 and AIME 2024–25 are near ceiling; MathArena lists AIME 2026 as deprecated; GPQA Diamond sits in the nineties for every frontier model (slide 23).

Harness sensitivity: one model's ARC-AGI-3 score was reported at 7.8, 17.8 and 38.3 across harness changes; HLE moves 4–13 points with tools. Compare within one harness, one effort.

Where discrimination is left: Humanity's Last Exam (best 67.7% with tools), ARC-AGI-3, Terminal-Bench, SWE-bench Pro, BrowseComp: agentic, long-horizon, tool-using.

Contamination: competition problems leak within weeks; report dates and decontamination.

Where the chains break: brittle, jagged, confidently wrong

The same problem in a different coat, a harness limit mistaken for a ceiling, and calibration that cuts both ways.

up to 65%
GSM-NoOp drop

33 vs 16%
hallucination o3 / o1†

>85%
confident and wrong

33 / 36
calibration wins

Brittleness: surface form and harness

GSM-Symbolic (Mirzadeh et al. 2024): change the names and numbers in GSM8K and scores fall and scatter; add one irrelevant clause (GSM-NoOp) and accuracy drops by up to 65%: the model subtracts the extra quantity. GSM1k: fresh problems, drops of up to 8% for some families.

Puzzles under a budget (Shojaee et al. 2025, "The Illusion of Thinking"): accuracy collapses beyond a complexity and chains shorten. Lawsen's comment: the collapse sits at the output-token limit, one puzzle is unsolvable as posed, and asking for a generating function instead of the move list succeeds.

Wrong with confidence

Hallucination: o3's system card reported 33% hallucinated claims on PersonQA against 16% for o1†: more claims of every kind. GPT-5 later claimed "about six times fewer" factual errors than o3†.

Calibration cuts both ways: Yoon et al. 2025: confidence better expressed in 33 of 36 settings; Mei et al. 2025: confidence above 85% on wrong answers, worse with longer chains; Zeng et al. 2025: fewer "I don't know"s, a reasoning tax. Humanity's Last Exam reports calibration errors of 81–93% for the 2025 models.

Safety: the chain as a window, and how it closes

A legible chain lets a monitor catch misbehaviour; optimizing against the monitor, or against length, hides it again.

the monitoring loop (Baker et al. 2025)



What the chain lets you see (2025)

Baker et al. (OpenAI): a GPT-4o monitor reading o3-mini's chain catches reward hacking in agentic coding with 95% recall, 60% from actions alone; training against the monitor keeps the hacks and hides them (recall near zero). Their proposal: a *monitorability tax*, the chain stays out of the loss.

Korbak et al. (41 authors): "a new and fragile opportunity"; Guan et al. (Dec 2025): 24 environments, models "fairly, but not perfectly, monitorable".

The rule: evidence while legible, never a guarantee.

How the window closes (2026)†

Faithfulness (slide 7): hints verbalized 25–39% of the time; reward hacks exploited on >99% of examples and mentioned in <2% of chains.

Control: OpenAI's CoT-Control study (Mar 2026) asked models to steer their own chain content on request: 0.1–15.4% success at baseline, falling with further RL; the chain is not yet a channel the model writes to deliberately.

Regression: the GPT-6 Astra system card (Sep 2026) reports monitorability below its predecessors and evaluation awareness at 9.6% of chains against 2.8% for GPT-5.6 Sol; its authors write that they "take the trend seriously".

Hidden by design: products show summaries or nothing; APIs return signed or encrypted thinking blocks that only the vendor's monitor reads.

What is settled and what is contested, October 2026

Two years of results sort into things every lab now relies on and questions the papers still argue about.

Settled enough to build on

1. Generated tokens add serial computation the single pass lacks; the circuit-complexity results and the benchmarks agree (slides 5–6).

2. RL on checkable answers with a group baseline turns a strong base model into a long, self-correcting reasoner in thousands of steps (slides 11–14).

3. The behavior copies: about a thousand curated traces move a 32B model to o1-preview level, and distillation beats RL on small models (slide 17).

4. Accuracy is log-linear in thinking tokens until it saturates; budgets and effort levels are the product interface (slides 19, 23).

5. On closed-form math and code the models exceed nearly all humans; the open problems are agentic and unverifiable (slide 25).

Still argued

Does RL add capability or sharpen it? Base models win pass@k at large k (Yue et al.); prolonged and broadened RL claims to move the boundary (ProRL, BroRL); a thousand steps and a Qwen base look alike (slide 16).

What does the chain say about the computation? Faithfulness 25–39%; monitorability measurable and regressing (slides 7, 27).

Beyond verifiable domains? Rubric rewards, self-certainty (Intuitor), self-proposed tasks (Absolute Zero): early and Qwen-heavy evidence.

How far does test-time compute go? Saturation bounds for parallel and serial scaling; 7.6M tokens per problem to match oracle pass@16 at pass@1 (Zhang et al. 2026).

How much is the base model? Random rewards work on Qwen and not elsewhere; one example suffices (slide 15).

In practice: using one, training one, evaluating one

Rules that follow from the lecture; the numbers are orders of magnitude, from the papers and cards cited.

Using a reasoning model

Set effort per task, not per deployment: start low and raise it only where measured accuracy moves (slide 19). Cap budgets; a hard problem has a latency floor and the chain is billed as output (slide 22). Use parallel samples only with a checker or a vote that makes sense (slides 9, 21). Keep previous chains out of the next prompt unless the API preserves them for you. Give the model tools rather than asking it to simulate them (slide 21). In agents, watch for deliberation replacing action (slide 20). Treat the visible chain as a summary, not a log (slide 27).

Training one

Start from a base with math and code in pretraining; its priors decide what RL can elicit (slide 15). Supervised fine-tune on 1k–100k long traces from a stronger model (OpenThoughts recipe; QwQ or R1 as teacher), then RL with GRPO or DAPO on 10k–100k problems with a checker: drop problems the model always or never solves, clip-higher, no KL term, token-level loss, overlong shaping, 8–16 rollouts of 8–32k tokens (slides 13, 18). Stacks: verl, OpenRLHF, TRL, SkyRL (slide 24). Budgets: $30 for a 3B toy, about $1k for a 32B SFT run, $300k for an R1-scale RL phase[†].

Evaluating one

Report pass@1 as the mean over 16–64 samples at a stated temperature, with the token budget and the harness; single runs on 30 AIME problems are noise. Report tokens per answer beside accuracy; a frontier, not a point (slide 20). Use fresh problems with dates (MathArena, LiveCodeBench) and perturbed forms (GSM-Symbolic) to catch contamination and brittleness (slides 25–26). Compare only within one harness and one effort level. Check what the chain omits: hint tests, reward-hacking probes, calibration on wrong answers (slides 7, 26–27).

Summary: what to take from this lecture

A reasoning model is a training recipe and an inference budget on the same transformer; both are now public.

Eight things to remember

1. The thinking span is generated text: serial compute, billed as output, usually hidden.
2. Prompting showed the effect; search showed inference compute helps if you can pick the winner.
3. RL on checkable answers removed the picking: one long chain, trained to check itself.
4. GRPO is REINFORCE with a group baseline; the 2025 fixes concern length and clipping, not the idea.
5. The base model supplies the behaviors; RL sharpens them; a thousand traces copy them.
6. Accuracy is log-linear in tokens, then saturates; effort is the knob you pay for.
7. The chain is evidence, not a transcript.
8. Benchmarks saturate in months; report dates, harness, tokens.

Open questions, October 2026

Is there a reward for problems without a checker? Rubrics, self-certainty and learned verifiers each work somewhere; none has the RLVR record.

Will the window stay open? Monitorability is in model cards and already regressing; latent reasoning (Coconut, recurrent depth) would close it by construction.

What is the unit of scaling? Training compute, rollouts, tokens per answer and parallel threads trade against each other; the exponents are being measured.

Who verifies the verifier? Models game verifiers, labs report their own scores; competition grading is the neutral ground.

The lecture notes expand every slide, with exercises and a paper kit of about 140 sources.

Sources

Papers by thread, then vendor material and community posts; † marks vendor, press and community sources.

Papers

Chains and theory: Nye et al. 2021; Wei et al. 2022; Kojima et al. 2022; Wang et al. 2022; Zhou et al. 2022; Madaan & Yazdanbakhsh 2022; Prystawski et al. 2023; Feng et al. 2023; Merrill & Sabharwal 2023; Li et al. 2024; Zelikman et al. 2024 (Quiet-STaR); Hao et al. 2024 (Coconut); Geiping et al. 2025. Faithfulness and monitoring: Turpin et al. 2023; Lanham et al. 2023; Arcuschin et al. 2025; Chen et al. 2025; Baker et al. 2025; Korbak et al. 2025; Guan et al. 2025; Helff et al. 2026. Search and verifiers: Cobbe et al. 2021; Uesato et al. 2022; Lightman et al. 2023; Yao et al. 2023; Besta et al. 2023; Wang et al. 2023 (Math-Shepherd); Zhang et al. 2024 (GenRM); Zhang et al. 2025 (PRM lessons); Snell et al. 2024; Brown et al. 2024; Guan et al. 2025 (rStar-Math); Liu et al. 2025; Fu et al. 2025 (DeepConf); Wen et al. 2025; Zheng et al. 2025 (Parallel-R1); Ghosal et al. 2025; Zhang et al. 2026. RL: Zelikman et al. 2022 (STaR); Singh et al. 2023; Shao et al. 2024 (DeepSeekMath); Lambert et al. 2024 (Tülu 3); DeepSeek-AI 2025 (R1; *Nature* 645:633); Kimi Team 2025 (k1.5, K2); Yu et al. 2025 (DAPO); Liu et al. 2025 (Dr. GRPO); Yue et al. 2025 (VAPO); Zheng et al. 2025 (GSPO); Hu et al. 2025 (REINFORCE++); Ahmadian et al. 2024 (RLOO); Yue et al. 2025 (pass@k); Shao et al. 2025 (spurious rewards); Wang et al. 2025 (one example); Liu et al. 2025 (ProRL); Hu et al. 2025 (BroRL); Cui et al. 2025 (entropy); Wang et al. 2025 (fork tokens); Zhao et al. 2025 (Intuitor; Absolute Zero); Huang et al. 2025 (rubrics); Khatri et al. 2025 (ScaleRL); Tan et al. 2025; Skalse et al. 2022; Wei et al. 2025 (SWE-RL); Feng et al. 2025 (ReTool); Gou et al. 2023 (ToRA); Chen et al. 2025 (pass@k training); Zhai et al. 2026; Cheng et al. 2026.

Data and distillation: Hsieh et al. 2023; Muennighoff et al. 2025 (s1); Ye et al. 2025 (LIMO); Guha et al. 2025 (OpenThoughts); Moshkov et al. 2025; Wen et al. 2025 (Light-R1); Chen et al. 2025 (AceReason). Length: Chen et al. 2024; Wang et al. 2025; Aggarwal & Welleck 2025 (L1); Hou et al. 2025; Ma et al. 2025 (NoThinking); Zhang et al. 2025 (AdaptThink); Fang et al. 2025 (Thinkless); Cuadron et al. 2025; Aghajohari et al. 2025; Sui et al. 2025 (survey). Limits and evaluation: Mirzadeh et al. 2024; Zhang et al. 2024 (GSM1k); Shojaee et al. 2025; Lawsen 2025; Yoon et al. 2025; Mei et al. 2025; Zeng et al. 2025; Hendrycks et al. 2021; Rein et al. 2023; Phan et al. 2025; Glazer et al. 2024; Chollet et al. 2025; Jain et al. 2024; Balunović et al. 2025. Model reports: Qwen3 2025; gpt-oss 2025; Phi-4-reasoning 2025; Llama-Nemotron 2025; MiniMax-M1 2025; GLM-4.5 2025; Magistral 2025. Surveys: Xu et al. 2025; Kumar et al. 2025; Lambert, *RLHF Book* ch. 7.

Vendor and community material†

OpenAI: Learning to Reason with LLMs (Sept 2024); Competitive Programming with LRMs (2025); o3 system card; CoT-Control (Mar 2026); GPT-6 Astra system card (Sep 2026); gpt-oss model card. Anthropic: extended-thinking and adaptive-thinking docs; Claude Opus 5.5 and Fable 5.1 cards. Google: Gemini thinking docs; Deep Think IMO post (Jul 2025). DeepSeek V4, Qwen3.8, Kimi K3 model cards; Hugging Face Open-R1 posts and test-time-compute post; NovaSky Sky-T1; Bespoke-Stratos; Agentica DeepScaleR; TinyZero; MathArena; ARC Prize 2025 report; *Scientific American* on R1's cost; AFP on IMO 2026; vendor pricing pages, all read Oct 21–22, 2026. Full list with arXiv identifiers in the paper kit.